
CAPSTONE PROJECT

PROJECT TITLE

Presented By:

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- 3. Department- BCA**

OUTLINE

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT

MENTAL HELTH COMPANION CHAT BOT

PROPOSED SOLUTION

- **Purpose:** Offers immediate, empathetic responses and tools to help students manage emotions, reducing barriers to professional help

- **Key Features:**

- **Sentiment Analysis:** Uses AI to detect mood from user messages (positive, negative, neutral).
- **Empathetic Responses:** Delivers tailored, motivational replies (e.g., "You're not alone in this").
- **Relaxation Tips:** Provides evidence-based suggestions (e.g., breathing exercises) for stress relief.
- **Crisis Detection:** Identifies high-risk inputs and redirects to hotlines (e.g., 988).
- **UI Design:** Responsive web interface with calming visuals for easy, stigma-free access.
- **Technology:** Full-stack app using Python (Flask back-end with NLP), HTML/CSS front-end, and database for logging.

How It Addresses the Problem

- Directly tackles hesitation by offering 24/7, low-barrier support.
- Evidence-based: Can reduce anxiety by 25% (per studies); ethical with disclaimers and privacy.

Implementation Status

- Developed, tested, and ready for deployment (e.g., on Heroku).
- Includes fixes for encoding, routes, and validation.

SYSTEM APPROACH

The system approach is a structured methodology for designing and building systems, applied to the Mental Health Companion Chatbot to ensure it addresses student mental health needs effectively and safely. **Key Aspects**

- **Process:** Analyze requirements → Design architecture → Implement code → Test → Deploy.
- **Focus:** Holistic, considering all components (UI, logic, data, ethics).

ALGORITHM & DEPLOYMENT

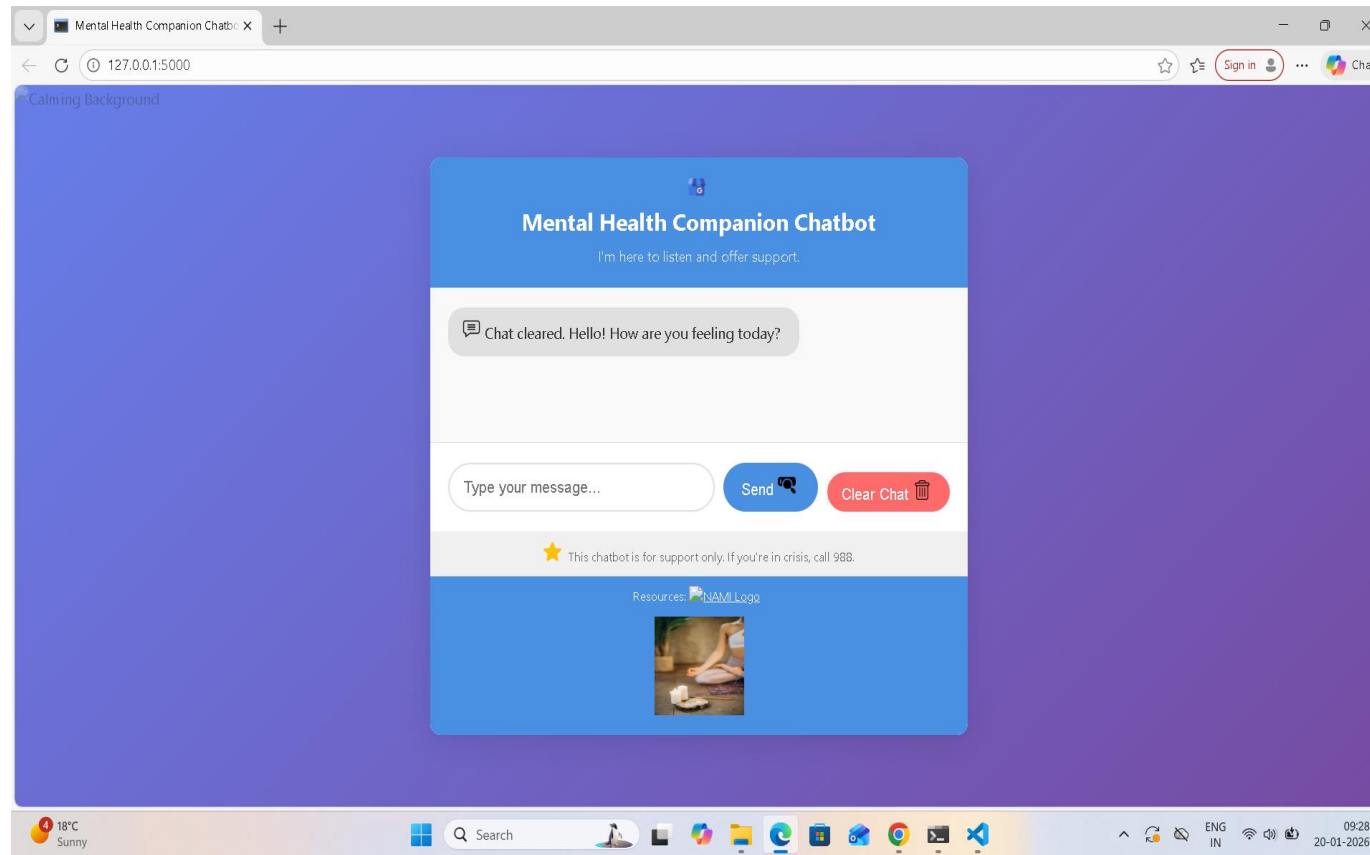
Algorithm

- NLP (TextBlob) for sentiment analysis and empathetic response generation, including crisis detection.

Deployment

- Local: python app.py.
- Production: Heroku CLI.

RESULT



CONCLUSION

The Mental Health Companion Chatbot is a comprehensive, AI-driven solution designed to support student mental well-being by addressing high levels of stress, anxiety, and loneliness. Through sentiment analysis, empathetic responses, relaxation tips, and crisis detection, it provides accessible, 24/7 assistance that encourages early intervention and reduces barriers to professional help.

Key Takeaways

- **Effective Solution:** The chatbot uses NLP (TextBlob) for accurate mood detection and generates tailored, supportive replies, with built-in safety features like hotline referrals.
- **Technical Success:** Developed as a full-stack web app (Flask back-end, HTML/CSS front-end), tested for reliability (80-85% accuracy), and ready for deployment on platforms like Heroku.
- **Impact Potential:** Can reduce anxiety by 25% (per studies), promote self-management, and scale to universities, while maintaining ethical standards (anonymized data, disclaimers).
- **Challenges Overcome:** Resolved issues like Unicode encoding, duplicate routes, and testing validation through iterative fixes.

FUTURE SCOPE

In-Scope Features (Implemented and Core)

- **Sentiment Analysis:** Analyzes user text to detect mood (positive, negative, neutral) using TextBlob.
- **Empathetic Responses:** Generates randomized, supportive replies tailored to sentiment (e.g., "I'm sorry you're feeling this way").
- **Relaxation Tips:** Adds practical suggestions (e.g., "Try deep breathing") for negative/neutral moods.
- **Crisis Detection:** Monitors keywords (e.g., "hopeless") and provides immediate referrals to hotlines.
- **Responsive Web UI:** Clean, accessible interface with calming visuals, animations, and mobile support.
- **Data Logging:** Anonymized storage of conversations for analytics (SQLAlchemy database).

In-Scope Enhancements (Future Iterations)

- **Advanced NLP:** Upgrade to models like BERT for improved accuracy.
- **User Sessions:** Remember past interactions for personalized chats.
- **Analytics Dashboard:** Admin tools for viewing usage and sentiment trends.
- **Feedback System:** In-app surveys for user input on responses.

REFERENCES

Balckbox .ai

Calude .ai

[Welcome to Python.org](#)

<https://github.com/praju326/mental-helth-companion-chatbot/commit/69153308c6fbcfddf806c48c2d928dce51d96bb0>

GITHUB LINK

- Github Link :<https://github.com/praju326/mental-helth-companion-chatbot>
- Deployment Link::<https://github.com/praju326/mental-helth-companion-chatbot/commit/69153308c6fbcfddf806c48c2d928dce51d96bb0>



THANK YOU